Mengqing Xu

E-mail: xumq2025@mail.sustech.edu.cn | Tel: (+86) 155-2737-1630 | Google Scholar

Education

Southern University of Science and Technology - Shenzhen, China Sep. 2025 – Jun. 2028
M.Sc. in Computer Science and Technology; Advisor: Prof. Pietro S. Oliveto

China University of Geosciences (Wuhan) - Wuhan, China Sep. 2021 – Jun. 2025
B.Eng. in Computer Science and Technology; GPA: 86/100

Research Profile

M.Sc. student at SUSTech, advised by Prof. Pietro S. Oliveto, working on the **runtime analysis of randomized search heuristics** and **evolutionary algorithms**. My current research focuses on designing hyper-heuristics and establishing rigorous runtime guarantees for their performance. I have contributed the main theoretical analyses and proofs to work accepted by IEEE TEVC and PPSN 2026, with a further manuscript under review. I am seeking Ph.D. opportunities in **theoretical computer science**, particularly in the areas of **randomized search heuristics** and **stochastic optimization**.

Publications

Author names are listed in Alphabetical Order, following the convention in theoretical computer science.

[1] Pietro S. Oliveto, **Mengqing Xu**. (2026). Automatic Adaptation of Mutation Rates for Move Acceptance Hyper-Heuristics Using Hypermutations. ***IEEE Transactions on Evolutionary Computation (IEEE TEVC).***

[2] Pietro S. Oliveto, Zhenyu Wang, Peizhou Wu, **Mengqing Xu**. (2026). On the Runtime Analysis of Reinforcement Learning Hyper-Heuristics. ***PPSN 2026.***

[3] Pietro S. Oliveto, Zhenyu Wang, Peizhou Wu, **Mengqing Xu**. Reinforcement Learning Hyper-Heuristics Using an Adaptive Penalty Parameter Can Achieve Optimal Runtime on Pseudo-Boolean Functions. ***AAAI 2027 (under review).***

Research Experience

Runtime Analysis of Complete Hyper-Heuristics Jan. 2025 – Jun. 2026
Primary Contributor | Advised by Prof. Pietro S. Oliveto

- Designed the Fast Adaptive MAHH_β, a complete hyper-heuristic combining heuristic selection and move acceptance; led the theoretical analysis, experimental evaluation, and first manuscript draft.
- Established the first rigorous runtime analysis of a complete hyper-heuristic using both heuristic selection and move acceptance, with guarantees across pseudo-Boolean benchmark functions.
- **Outcome:** Accepted for publication in *IEEE Transactions on Evolutionary Computation (IEEE TEVC), 2026.*

Runtime Analysis of Reinforcement Learning Hyper-Heuristics Sep. 2025 – Present
Primary Contributor | Collaborating with Zhenyu Wang and Peizhou Wu, advised by Prof. Pietro S. Oliveto

- Established the first rigorous positive result for a hyper-heuristic using a reinforcement learning mechanism from the literature, proving that, with suitable parameters, RLHH optimizes LeadingOnes in the best possible expected runtime achievable with RLS_1 and RLS_2, up to lower-order terms; led the runtime analysis and first manuscript draft.
- Proposed and rigorously analyzed an adaptive penalty mechanism that replaces a fixed, problem-dependent penalty with an online recalibration scheme; proved optimal runtimes on LeadingOnes and OneMax, including the first hyper-heuristic result matching the unary unbiased black-box complexity of OneMax up to its leading constant; led the runtime analysis and first manuscript draft.
- **Outcomes:** Published in *19th International Conference on Parallel Problem Solving from Nature (PPSN 2026)*; follow-up manuscript submitted to AAAI 2027 and currently under review.

Teaching Experience

Teaching Assistant, CS217: Data Structures and Algorithms Analysis (Honors), SUSTech Fall 2025; Fall 2026
Teaching Assistant, CS338: Introduction to Theory of Computation, SUSTech Spring 2026

Honors and Awards

- Top-tier Graduate Academic Scholarship and Research Assistantship Award, SUSTech (Top 15%) 2026
- Innovation and Entrepreneurship Scholarship, China University of Geosciences (Wuhan) 2022
- Third Prize, National College Student Mathematical Modeling Competition 2022
- Second Prize, Huazhong Cup Mathematical Modeling Competition 2021

Academic Activities

- Presenter, 19th International Conference on Parallel Problem Solving from Nature — Trento, Italy, 2026.

Skills

- **Mathematics:** Probability theory, stochastic processes, drift analysis.
- **Programming:** Python and LaTeX.
- **Languages:** Chinese (Native); English (Fluent).